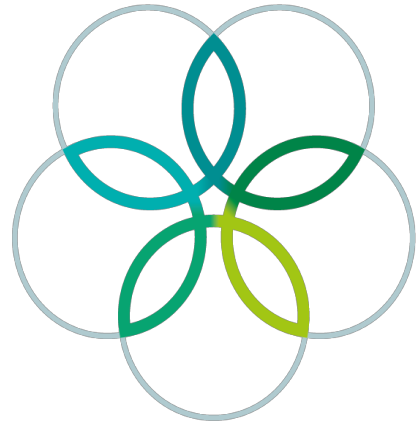
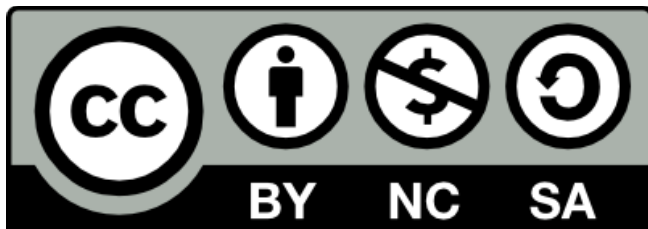


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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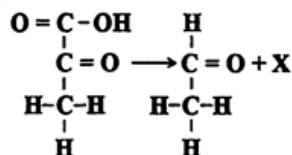
II INTERNATIONAL BIOLOGY OLYMPIAD

1. How many triplets of DNA nucleotides code 20 amino acids?

- a) 20; b) 61; c) 64; d) 4.

2. The above reaction is catalysed by:

- a) dehydrogenase;
b) decarboxylase;
c) oxidase;
d) hydrolase.



3. The function of the so-called "nonsense" codons is:

- a) to code each one for several amino acids;
b) to enable the mRNA to attach itself to any ribosome;
c) to code each one for the proper amino acid;
d) to indicate the end of the coded message in the mRNA.

4. A peptide bond is formed between:

- a) phosphate and carboxyl; b) carboxyl and amino;
c) alcohol and aldehyde; d) aldehyde and amino.

5. Cellulose and glycogen are both:

- a) polysaccharides; b) glucose and glucose-1-phosphate polymers;
c) polynucleotides; d) disaccharides.

6. The sequence of nucleotides of mRNA is complementary to the sequence of nucleotides of:

- a) the 2 strands of a DNA molecule; b) one of the strands of a DNA molecule;
c) the molecule of tRNA; d) all the tRNA molecules.

7. The DNA triplet GCT is complementary to the tRNA anticodon:

- a) GCT; b) CGA; c) UGC; d) GCU; e) GCT.

8. Bacteria cultured in an environment containing light nitrogen isotopes (^{14}N) were transferred to an environment containing heavy nitrogen (^{15}N) for a period corresponding to one replication and then they were brought back to the initial environment. The analysis of bacteria DNA composition made after a period corresponding to two successive replication showed that correlation of $^{14}\text{N}/^{15}\text{N}$ in DNA was:

- a) 1:1; b) 2:1; c) 4:1; d) 8:1.

9. All procaryotic cells have:

- a) ribosomes and mitochondria;
b) a plasma membrane and vacuoles;
c) a plasma membrane and nuclear membrane;
d) a plasma membrane and ribosomes.

10. Any kind of cell has a plasma membrane built of:

- a) lipids and proteins; b) lipids only;
- c) proteins only; d) lipids and polysaccharides.

11. At the time when a cell bagging to divide, it has N chromosomes and Q DNA.

What DNA quantity and chromosome number is expected in each daughter cell after mitosis?

- a) N and Q; b) N/2 and Q/2; c) N and Q/2; d) N/2 and Q.

12. Molecules of water get to the leaves by the following way:

- a) stomata → mesophyll → xylem; b) xylem → mesophyll → stomata;
- c) phloem → xylem → mesophyll; d) phloem → mesophyll → stomata.

13. Meiosis produces:

- a) gametes in all cases; b) spores in all cases;
- c) spores and gametes in both animals and plants;
- d) usually spores in plants and gametes in animals.

14. Antibodies are synthesised:

- a) in macrophages only; b) in lymphocytes only;
- c) in macrophages and lymphocytes; d) in macrophages and neutrophils.

15. Secretion of the follicle-stimulation hormone (FSH):

- a) affects the ovary only; b) stimulates follicles growth in the ovary;
- c) is controlled by the hypothalamus and the ovary;
- d) is not known for man.

16. A boy belongs to the 0 blood group, his mother - to the A blood group, and his father - to the B blood group. The probability of his sister belonging to the same type is:

- a) 1/16; b) 1/8; c) 1/4; d) 1/2.

17. Autosomes are:

- a) nonsex-chromosomes; b) cells which can reproduce themselves;
- c) organelles which can reproduce themselves; d) sex-chromosomes.

18. The certain species of fungus grow on animal facals. How this type of feeding is called?

- a) parasitism; b) commensalism; c) saprophytes; d) symbiosis.

19. The human forearm and the bird's wing have the same structure. This suggests that:

- a) birds are derived from mammals;
- b) birds and mammals have common ancestors;
- c) mammals are less adapted to their environment than birds;
- d) birds and mammals are well adapted for flying.

20. People with type 0 blood are:

- a) universal donors; b) universal donors and recipients;
- c) universal recipients; d) unable to give their blood for any blood transfusion.

21. Insulin injecting into human blood:

- a) reduces glycogen synthesis; b) increases muscles consumption of glucose;
- c) reinforces glycogen destruction; d) increases the blood glucose content.

22. Lung ventilation is necessary to maintain concentration of carbon dioxide and oxygen of the following relation:

- CO_2 O_2
a) high — high;
b) low — low;
c) low — high;
d) high — low.

23. In a plant cell the organelles separated by two membranes from cytoplasm are:

- a) the nucleus only; b) the mitochondria and the plastides only;
c) the nucleus, the mitochondria and the plastides;
d) the mitochondria, the lysosomes and the plastides.

24. What days of female menstrual cycle fertilization is mostly probable?

- a) the 7-th — the 9-th; b) the 8-th — the 10-th;
c) the 11-th — the 17-th; d) the 21-th — the 28-th.

25. Sexual reproduction in plants and animals:

- a) increases variation; b) reduces variation;
c) is able to promote or reduce variation; d) has no effect on variation.

26. Cells from the skin of a person suffering from Down's syndrome are examined through the microscope. Which of the following could be observed in each cell?

- a) additional chromosome;
b) the shortage of chromosomes;
c) a haploid set of chromosomes;
d) chromosome number differs from that in adjacent cells.

27. Human individual features depend:

- a) on genotype entirely; b) on environment impact entirely;
c) on interaction between genotype and environment;
d) on parents genotype entirely.

28. DNA composition analysis showed that one of the listed correlations cannot change:

- a) A/T ; b) G/C ; c) $A+T/G+C$; d) $A+G/T+C$.

29. Bacteria are characterised by the following peculiarity:

- a) they have a cell structure; b) they are diploid;
c) they can be seen through the powerful microscope;
d) they are free living, parasitic or symbiotic organisms.

30. Which of the listed below organs produce non-active enzyme forerunner into mammal's digestive tract?

- a) salivary glands; b) pancreas; c) gall bladder; d) liver.

31. Which of the listed below processes is not connected with adrenaline action?

- a) stimulation glycogen change to glucose;
b) speeding up the heart contractions;
c) intensification peristalsis of the intestine;
d) pupil's dilating.

32. A simple key for determination four different plants is worked up:

1. The vascular tissues are present.....2;

The vascular tissues are absent.....Plant A.

2. Male gametes have flagellums.....3;

Male gametes don't have flagellums.....Plant B.

3. Spores of one type.....Plant C;

Spores of two type.....Plant D.

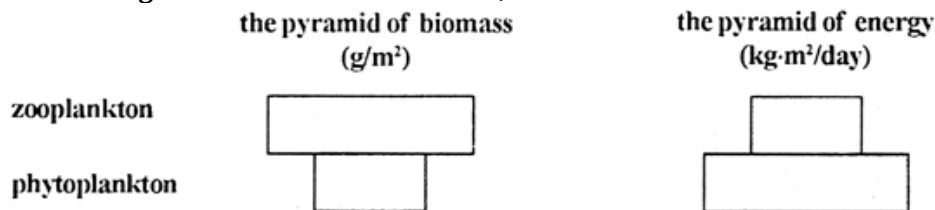
Which of this plants is an angiosperm one?

a) A; b) B; c) C; d) D.

33. The sporophyte generation of the flower plant is a diploid one. Then its endosperm is:

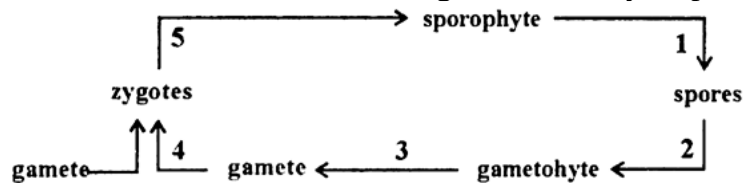
a) n ; b) $2n$; c) $3n$; d) $4n$.

34. The diagrams illustrate the situation, when:



- a) zooplankton reproduces faster, than phytoplankton;
- b) one zooplankton generation feeds on one phytoplankton generation;
- c) one phytoplankton generation develops food for several zooplankton generations;
- d) one zooplankton generation feeds on several phytoplankton generations.

35. The scheme illustrates some stages and processes of green plant life cycle. Which of the listed below pairs of processes are able to make the most valuable contribution to the genetic diversity of species?



- a) 1 and 2;
- b) 1 and 4;
- c) 2 and 5;
- d) 3 and 4.

36. Which of the soil processes is useless for vascular plants?

- a) nitrogen fixation from atmosphere; b) oxidation ammonium compounds;
- c) oxidation nitrites into nitrates; d) reduction nitrates to nitrogen.

37. Vitamin, which is the most important for growth and crepuscular vision, is:

a) A; b) B; c) C; d) D.

38. Breathing is more rapid, when physical load is concerned with?

- a) high O₂ blood concentration; b) low O₂, blood concentration;
- c) high CO₂ blood concentration; d) low CO₂ blood concentration.

39. The results of blood testing are represented in the table:

Plasma	Agg	utinant	Antibodies Rh ⁺
	a	b	
Agglutination	-	+	-

What blood was taken for analysis?

- a) A, Rh⁺; b) B, Rh⁺; c) B, Rh⁺; d) AB, Rh⁺.

40. While breathing:

- a) the diaphragm contracts; b) intercostal muscles contract;
c) abdominal muscles contract; d) no muscles contract.

41. The nerve impulse travels in the organism by one of the listed below ways:

- a) dendrite → synapse → neuron body → axon;
b) axon → neuron body → dendrite → synapse;
c) dendrite → neuron body → axon → synapse;
d) dendrite → synapse → axon → neuron body.

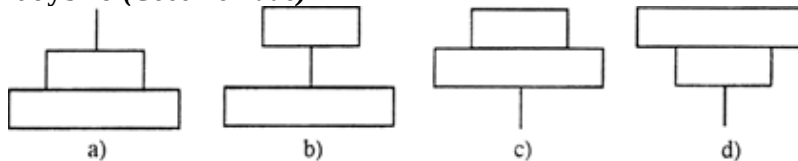
42. Transfer of Rh is:

- a) transfer of genetic information from DA/A to mRNA;
b) transfer of information from mRNA to rRNA;
c) organition of amino acids according to the codons;
d) recognition of anticodons mRNA.

43. Which of J.B. Lamarck's statements is turned down nowadays?

- a) sometimes it is difficult to distinguish one species from another with the help o morphological features;
b) living beings vary; c) benefit features are heredited;
d) living beings adapt themselves to the environment.

44. Which of the number pyramids correlates with the food chain: plants-aphid (*Aphididae*) - ladybird (*Coccinellidae*)



45. Which of the listed birds will get more evolution benefit?

Bird	Number		
	laid eggs	hatched nestling	reproducing offspring
a)	9	8	2
b)	2	2	2
c)	9	9	3
d)	7	5	4

46. The general cause of verdure correlated variations from the South to the North and from the plain to the peaks of the mountains, i.e. with increasing latitude as well as altitude is connected with:

- a) while altitude as well as latitude increases temperature declines;
b) sun lights fall angle is smaller on mountain slopes;
c) clouds gather over the mountain peaks;
d) plants have some difficulties about slope setting.

47. In biocenosis nutritious substances cycling isn't concerned with:

- a) transference some food elements from an organism to atmosphere;
- b) joining the most food elements to food webs through animals;
- c) increasing of population density in that regions where food elements storage are more than in another;
- d) number limitation of ecosystem organisms caused by shortage of some food elements.

48. Which of the listed below factors affect is correlated with population number least of all?

- a) parasitism;
- b) accumulation of metabolism wastes;
- c) predatoriness;
- d) hard winter.

49. The population is able to increase its number exponentially:

- a) when only food is limited;
- b) when firstly it occurs in the surrounding proper for this species but not yet inhabit by it;
- c) only when predators are absent;
- d) in laboratory only.

50. The most effective action about improving the environment is:

- a) the second use of glass package;
- b) active participation in the movement for developing laws protecting the environment;
- c) bicycle instead of driving;
- d) become a vegetarian.

51. In the nucleus of rabbit's cell haploid DNA number weighs $4 \cdot 10^{-n}$ g. What weight will zygote's DNA have in prophase of the first mitosis?

- a) $4 \cdot 10^{-12}$ g;
- b) $8 \cdot 10^{-12}$ g;
- c) $1,6 \cdot 10^{-11}$ g;
- d) $3,2 \cdot 10^{-11}$ g.

52. In what test does the *Escherichia coli* not discovering cause troubles?

- a) drinking water;
- b) newborn child fecal;
- c) adult fecal;
- d) adult urine.

53. In the desert the limited factor for plants is usually:

- a) duration of daylight;
- b) salinization of soil;
- c) quantity of moisture;
- d) temperature.

54. The most ancient paleontological illustrations of life on the Earth are dated:

- a) 32 million years;
- b) 600 million years;
- c) 3,5 milliard years;
- d) 4,5 milliard years.

55. In what part of human body discovering of bacteria *Staphylococcus epidermidis* does not cause a trouble?

- a) skin surface;
- b) nasal cavity;
- c) abdominal cavity;
- d) vagina.

56. 200 amino acids form the protein. The gene specifying its structure:

- a) would be longer in a procaryote;
- b) would be longer in an eucaryote;
- c) would be same length in a procaryote and in an eucaryote.

57. A cilium differs from a flagellum by:

- a) its bases doesn't possess 9+2 arrangement of microtubules;
- b) flagellum is capable of rotatory motion whereas, cilium is not;
- c) cilia are longer;
- d) there is no difference.

58. How many centrioles are seen in the beginning of prophase in human cell?

- a) 1; b) 2; c) 4; d) a lot.

59. How many chromosomes one can observe simultaneously when anaphase finishes in human skin cell?

- a) 23; b) 46; c) 69; d) 92.

60. Home fly (*Musca domestica*) is able to adapt to environmental changes quicker than human because:

- a) it is smaller; b) it is a good flyer;
c) it has a great number of offspring; d) rapid succession of generations occurs.

61. Lipid's binary membranes are not freely permeable to which of the following molecules?

- a) oxygen; b) carbon dioxide; c) amino acids; d) water.

62. The result of Na^+ - K^+ -pump function is not:

- a) low Na^+ concentration in the cell;
b) facilitated diffusion of amino acids into the cell;
c) developing the proton's gradient of concentration;
d) high K^+ concentration in the cell.

63. If one-cell animal (*Amoeba proteus*) and the erythrocyte are put into distillate water:

- a) both cells would be destroyed;
b) amoeba would be died, and the erythrocyte would go on living;
c) amoeba would go on living and the erythrocyte would be died;
d) both cells would go on living.

64. Which of the following ways of transporting molecules across membranes utilise ATP directly?

- a) diffusion; b) Na^+ - K^+ -pump;
c) facilitated diffusion; d) transporting across open channels.

65. Of the following crosses, which is a testcross?

- a) AA x aa; b) AA x Aa;
c) Aa x Aa; d) Aa x aa.

66. When a healthy mouse is infected with *Pneumococcus*, it is likely to become ill and die if the *Pneumococcus* is:

- a) heat-killed; b) is alive, but lacking a polysaccharide coat;
c) a mixture of some heat-killed *Pneumococcus* having polysaccharide coat and some *Pneumococcus* lacking coat;
d) both lacking polysaccharide coat and heat-killed.

67. The eucaryotic cell, with one more chromosome is:

- a) diploid; b) haploid; c) aneuploid; d) monoploid.

68. Who proposed a coherent theory in the field of evolution first:

- a) Charles Darwin; b) Jean Baptiste de Lamarck;
c) Alfred Wallace; d) Julian Kuxley.

69. In the rocks of nearly 3,5 milliard years the fossils are found of:

- a) viruses; b) dinosaurs; c) bacteria; d) algae.

70. The first fossils of genus *Homo* were found:

- a) in South Australia;
- b) in South Africa;
- c) in South-East Asia;
- d) in Europe.

71. Viruses can contain:

- a) only DNA; b) only RNA; c) DNA and RNA; d) DNA or RNA.

72. Charles Darwin's explanation of the way, how evolution occurs is that:

- a) God determines, which species should evolve;
- b) fitness enable one species to leave more offspring;
- c) certain species has "built" in plan of evolution;
- d) environmental impact causes adequate species adaptations.

73. The major factor, causing deviations from Hardy-Wainberg equilibrium is:

- a) mutations; b) migration; c) selection; d) near-related breeding.

74. First fossils of Australopithecines were found:

- a) in Africa; b) in Asia; c) in America; d) in Australia.

75. What feature is not peculiar to *Azotobacter*?

- a) procaryotic; b) chemosynthetic; c) aerobic; d) atmospheric nitrogen fixation.

76. For *Trypanosoma* the host, in which sexual reproduction occurs, is:

- a) human; b) fly; c) antelope; d) none of listed above.

77. Charles Darwin considered that the most doubt about the truth of his theory is:

- a) Earth age; b) lack of blanks in palaeontology history;
- c) ways of traits heredity; d) Charles Darwin had no doubts.

78. All *Fungi* are characterised by:

- a) spore reproduction; b) heterotrophic food;
- c) mycelium; d) developing basidiocarp.

79. Angiosperms dominate in the modern Earth flora because:

- a) they have well-developed tissues securing high metabolism;
- b) they have the organ of seed reproduction - the flower;
- c) they have C₃- and C₄-photosynthesis;
- d) a + b + c.

80. What plant cells can function after their death?

- a) cambium; b) sieve-tubes; c) collenchyma; d) vassels.

81. Malaria is caused by:

- a) mosquito bite (*Anopheles maculipennis*); b) moisture air of marshes;
- c) *Plasmodium vivax*; d) *Bacillus malaria*.

82. Cell walls of *Fungi* are mainly made of:

- a) glucose; b) cellulose; c) pectin; d) chitin.

83. One can consider the bryophytes as the dead-end evolutionary siding because:

- a) they have no true roots; b) their vascular tissues are badly developed;
- c) gametophyte dominates in the life cycle; d) sex reproduction depends on water.

84. You sneeze and have watery eyes every spring. Your physician tests you and says that you are allergic to pollen of a plant *Ambrosia*. You guess that *Ambrosia* is:

- a) high plant;
- b) insects pollinated;
- c) wind pollinated;
- d) excreta phytoncids.

85. All helminths are characterised by:

- a) the digestive system is absent;
- b) high reproduction intensity;
- c) the senses are absent;
- d) hermaphroditism.

86. There are 24 chromosomes in the cells of pine (*Pinus sylvestris*) leaves. Then its endosperm would contain chromosomes:

- a) 12 chromosomes;
- b) 24 chromosomes;
- c) 36 chromosomes;
- d) 48 chromosomes.

87. Which of the listed pairs aren't homologous organs?

- a) tendrils in grapes (*Vitis*) and tendrils in the garden pea (*Pisum*);
- b) tendrils in grapes (*Vitis*) and strawberry (*Fragaria*) runners;
- c) tendrils in grapes (*Vitis*) and hawthorn (*Crataegus*) prickles;
- d) needles (*Pinus*) and cactus prickles (*Mammillaria*).

88. The seed germinating on the soil surface has a root growing vertically nevertheless of seed's posture. It is because of:

- a) negative heliotropism;
- b) positive geotropism;
- c) substratum structure;
- d) positive hydrotropism.

89. Organic compounds pass through xylem:

- a) after autumn leaf-fall only;
- b) at night only;
- c) only when buds of leaf-fall trees develop;
- d) never.

90. Sponges (*Porifera*) digestion is:

- a) out intestine;
- b) in the cavity;
- c) in the cell;
- d) b + c.

91. The common features for *Annelida* and *Arthropoda* are:

- a) exoskeleton;
- b) having extremities;
- c) closed circulatory system;
- d) segmentation.

92. *Lymnaea stagnalis* - lung mollusc, inhabiting water permanently, lay his eggs:

- a) on above-water plant parts;
- b) into the bank soil;
- c) on underwater plant parts;
- d) digs into silt.

93. Flatworms (*Plathelminthes*):

- a) don't have any circulatory system and body cells get their feed and oxygen as a result of diffusion;
- b) have closed circulatory system and a heart;
- c) have opened circulators' system;
- d) don't need any circulatory system because they don't have any organs.

94. Air flows through birds lung:

- a) on inhalation from the tail to the head;
- b) on exhalation from the head to the tail;
- c) on inhalation and on exhalation from the tail to the head;
- d) on inhalation from the head to the tail and on exhalation from the tail to the head.

95. Sting of a bee (*Apis mellifera*) is:

- a) modified organ of copulation; b) modified abdomen segment;
- c) modified ovipositor; d) neither the one nor the others.

96. White bears (*Thalarctos maritimus*) don't eat penguins (*Impennes*) in wildlife because:

- a) they don't meet each other in wildlife;
- b) penguins swim faster than bears;
- c) penguins live within groups and it is dangerous to attack them;
- d) penguins meat is toxic.

97. Horsefly (*Tabanus*) females eat:

- a) blood; b) plant juice;
- c) excrements of hoofed animals; d) no one of the variants.

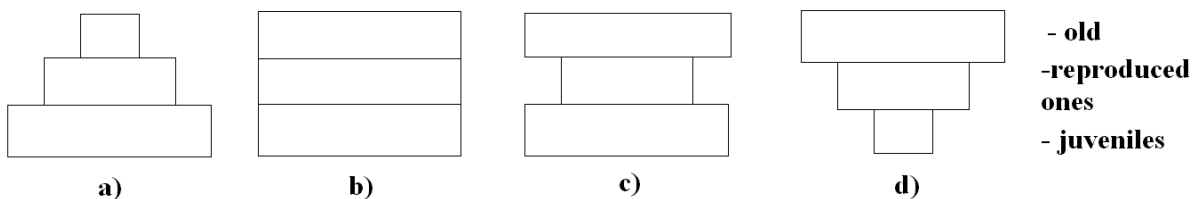
98. High radiation of human body doesn't cause:

- a) destroying of blood cells formation; b) cancer;
- c) stomach-intestine bleeding; d) myocard infarction.

99. There are not more than 6 links in food chains usually because:

- a) environmental resources are limited;
- b) large amount of energy is lost at every link;
- c) predators are unable to eat predators; d) a + b + c.

100. There is age distribution in a population in schemes. Which of the populations has a greater chance to extinct?



*****Following information deals with the questions № 101-135.
Decide true or false that or either statement.**

101. Glycocalyx (the highest layer of a great number of animal cells) contains different polysaccharides and proteins.

- a) yes; b) no.

102. Cell plasma membrane contains different proteins only.

- a) yes; b) no.

103. Water, ions of different salts, mineral and organic molecules pass through the narrow channels in the cell membrane from the environment into the cell.

- a) yes; b) no.

104. Pinocytosis is common for animal, fungi and bacteria cells only.

- a) yes; b) no.

105. Polysaccharides prevail in the basis matter of plant cells cytoplasm.

- a) yes; b) no.

106. When guard cells turgor is higher than usual, the stomata is opened.

- a) yes; b) no.

107. Only cell nucleus contains DNA, where it forms chromosomes.

a) yes; b) no.

108. Ameba, infusoria and great amount of other one-celled animals (*Protozoa*) feed by phagocytosis.

a) yes; b) no.

109. Phagocytosis is common for animal cells as well as for plant, bacteria and blue-green algae (*Cyanophyta*).

a) yes; b) no.

110. Proteins are synthesized on the membranes of smooth endoplasmic reticulum.

a) yes; b) no.

111. Ribosomes are common for cells of all living beings.

a) yes; b) no.

112. Ribosomes contain proteins, RNA, lipids and polysaccharides.

a) yes; b) no.

113. Mitochondria are found in cytoplasm of the most plant and animal cells.

a) yes; b) no.

114. Lysosomes are formed by the Golgi complex.

a) yes; b) no.

115. In all plant and animal cells there is an organelle located near the nuclear envelope and named cell center (centrioles).

a) yes; b) no.

116. RNA, DNA and proteins form nucleolus.

a) yes; b) no.

117. Nucleolus are the places where big and small ribosomes parts are formed.

a) yes; b) no.

118. Human arm and whale's (*Balaenoptera*) flipper are homologous organs.

a) yes; b) no.

119. Tendrils of the garden pea (*Pisum sativum*) and tendrils of the cucumber (*Cucumis sativus*) are analogous organs.

a) yes; b) no.

120. Barberry (*Berberis vulgaris*) prickles and dog-rose (*Rosa canina*) prickles are homologous organs.

a) yes; b) no.

121. Melanism appears as a result of genetic diversity and is enable to be "caught up" by natural selection.

a) yes; b) no.

122. Cross-pollination always gives more benefits than self-pollination.

a) yes; b) no.

123. Platelets are formed in the spleen.

a) yes; b) no.

124. Mitochondria were not found in the cells of some anaerobic organisms.

a) yes; b) no.

125. Animals adapt to sweet and salt substances better than to bitter and sour ones.

a) yes; b) no.

126. Some Australian marsupials (*Metatheria*) species and species of placental (*Eutheria*) mammals are convergent ones.

a) yes; b) no.

127. Tympanic membrane is common for every land vertebrate animal ears.

a) yes; b) no.

128. The first full conception about evolution was developed by J.Sant-Iler.

a) yes; b) no.

129. Natural selection is the only force that is capable to maintain the high μ level of heterozygotes in wildlife populations.

a) yes; b) no.

130. The term "genetics" was suggested by G.Mendel.

a) yes; b) no.

131. The Eustachian tube preserves the eardrum (tympanic membrane) from destroying when atmosphere pressure fluctuates.

a) yes; b) no.

132. Tadpole tails disappear because of dead cells digesting by lysosomes.

a) yes; b) no.

133. In cephalopod's eye accommodation achieves by changing the shape of the lens.

a) yes; b) no.

134. Penguins have a keel on the sternum.

a) yes; b) no.

135. Oviparous (*Prototheria*) occur not only in Australia and neighbour islands. They live in South America as well.

a) yes; b) no.

*****Following information deals with the questions № 136-165.**

Write the terms proceeding from the listed definitions.

The codes of the answers:	11. leucoplast;	21. recolonization;
1. hair cells of organ of Corti;	12. macroevolution;	22. coprophagous;
2. hemolymph;	13. mantle;	23. symbiote;
3. divergence;	14. mesoderm;	24. symplast;
4. intron;	15. migration;	25. sporophyte;
5. clone;	16. mycorrhiza;	26. thermophobic;
6. commensalism;	17. ontogeny;	27. tissue;
7. convergence;	18. vane;	28. evolution;

- | | | |
|-------------------|----------------------|--------------|
| 8. rhizome; | 19. parthenogenesis; | 29. neuston; |
| 9. crossing-over; | 20. pericardium; | 30. etology. |
| 10. xerophyte; | | |

136. The certain or by chance movement of organisms of some species from the area lines...

137. The permanent or temporary living together organisms of various species in which one of the partners eats meat residues or excretes of the other not harming it...

138. Vegetative genetically identical offspring of one organism...

139. An organism which eats excrements of other animals, mainly mammals..

140. A group of living beings inhabiting upper level of water on the border with air...

141. Artificial return to some territory of the species extinct there before...

142. Origin of living beings from former living forms...

143. Plants of arid habitants stable to overheating and dehydrating because of a number of adaptive features...

144. Colourless plastides of various form and function in plant cells...

145. Organism inhabiting at permanent low temperature conditions...

146. The exchange of corresponding chromatid segments between homologous chromosomes responsible for new combinations of various alleles...

147. Reproducing part of hearing system of mammals and humans, which transfer the energy of sound waves into the nervous impulse...

148. The independent development of similar structures in unrelated organisms to identical environment...

149. Interaction between members of the population of one species or various species in order to obtain a mutually required resource available in limited supply...

150. Variant of bone tissue forming the placoid fish scales and the main part of mammals teeth...

151. Liquid tissue circulating in insects body...

152. The type of evolution coming into existence of taxons higher than species...

153. The outer crease of the skin in *Mollusca*, *Brachiopoda*, *Cirripedia* covering the whole body or its part...

154. Embodiment of plant or animal cells in tissues while out-cells matter is destroying or dissolving...

155. A symbiotic association between fungi and the roots of a vascular plant...
156. Origin of the group of organisms from common ancestor...
138. Vegetative genetically identical offspring of one organism...
139. An organism which eats excrements of other animals, mainly mammals..
140. A group of living beings inhabiting upper level of water on the border with air...
141. Artificial return to some territory of the species extinct there before...
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153. The outer crease of the skin in *Mollusca*, *Brachiopoda*, *Cirripedia* covering the whole body or its part...
154. Embodiment of plant or animal cells in tissues while out-cells matter is destroying or dissolving...
155. A symbiotic association between fungi and the roots of a vascular plant...
156. Origin of the group of organisms from common ancestor...
- 46
157. The course of development of an individual, the whole complex of transformations

from egg to life end (death or the new division of an individual)...

158. Lamellate-like part of a contour feather growing from both sides of the shaft...

159. One of the forms of sex reproduction when female sex cells develop without fertilization...

160. Strong sac of connective tissue enclosing the heart of some invertebrate and all vertebrate animals...

161. Segments of the gene which has not genetical information...

162. Science about behaviour of living beings...

163. Unity of cells of multicellular organism (except sex ones)...

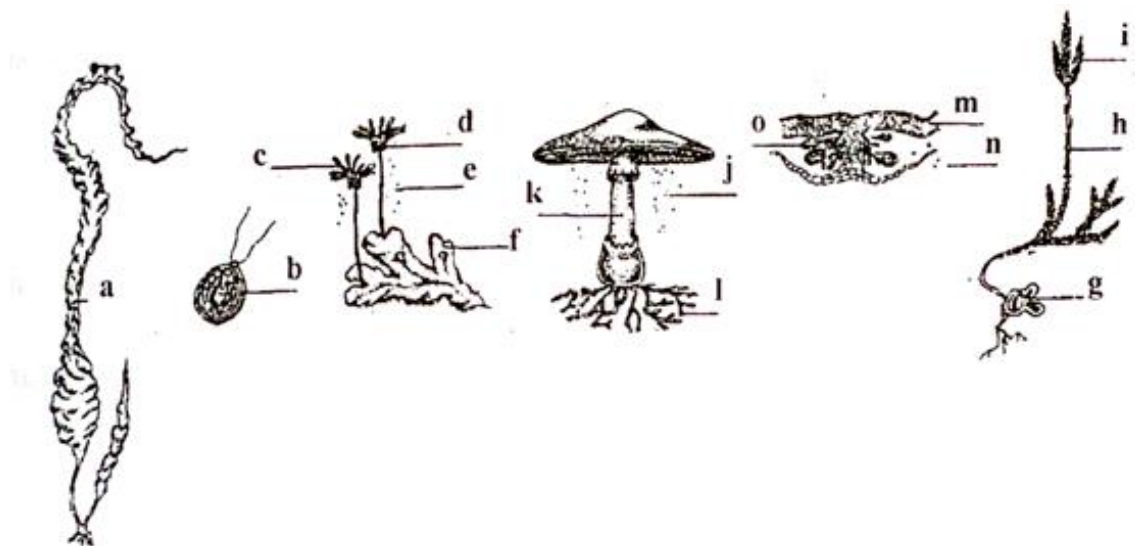
164. The asexual plant generation producing spores...

165. A modified shoot with long thin internodes and scale colourless, rarely green leaves...

*****Following information deals with the questions № 166-169.
Compliment the letters of the figures with codes of the answers.**

166. What is the chromosome number of the organisms and their generations, which are marked by letters in the figure?

The codes of the answers: 1. n ; 2. $n + n$; 3. $2n$; 4. $3n$.



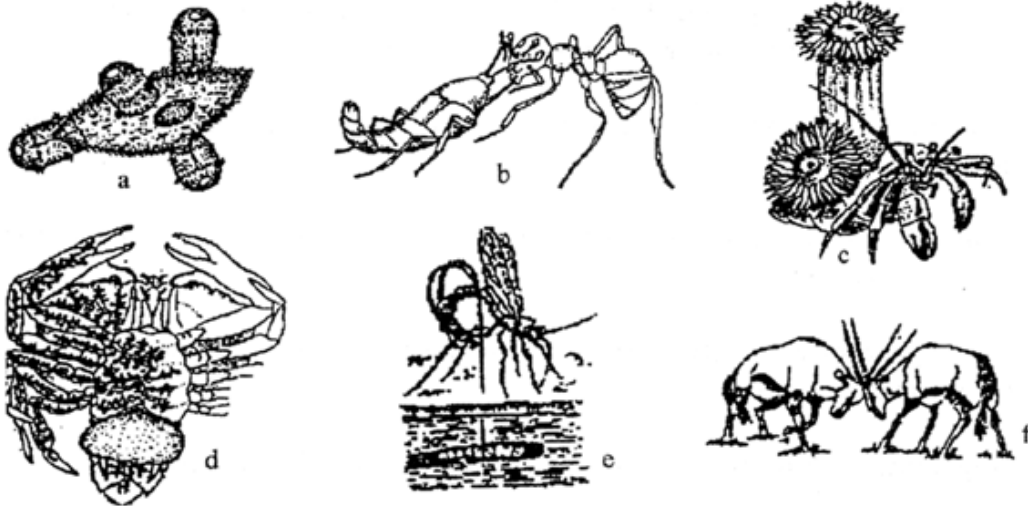
Answers:

a) b) c) d) e) f) g) h)
i) j) k) l) m) n) o)

167. What terms refer to types of interactions of the living organisms, which are shown in the figure?

The codes of the answers:

- | | | |
|------------------|----------------|-----------------|
| 1. neutralism; | 3. mutualism; | 5. predation; |
| 2. commensalism; | 4. parasitism; | 6. competition. |



Answers: a) b) c) d) e) f)

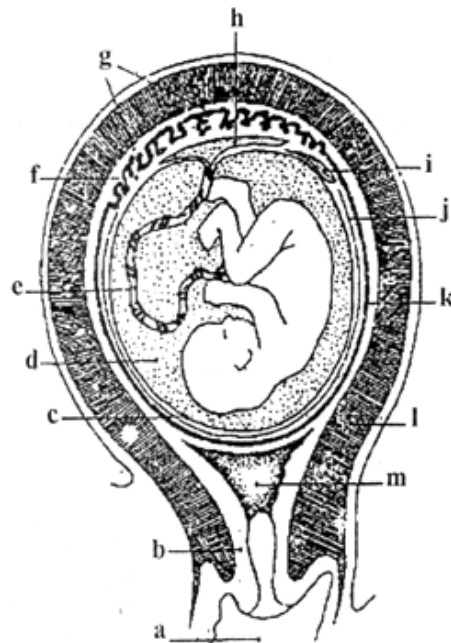
168. Find the connection between terms and structures, which are shown in the figure:

The codes of the answers:

1. uterus cavity;
2. cervix;
3. miometrium;
4. endometrium;
5. vagina;
6. chorion;
7. chorion villus;
8. placenta;
9. amnion;
10. amnion cavity;
11. umbilical cord;
12. allantois;
13. residues of yolk sac.

Answers:

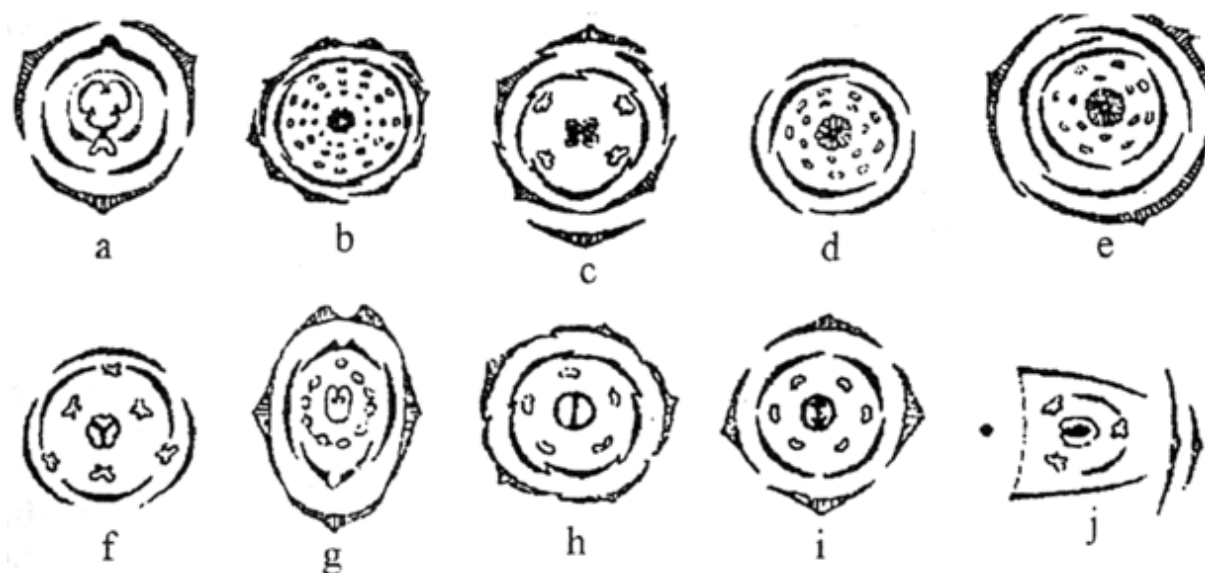
a) b) c) d) e) f)
g) h) i) j) k) l) m)....



169. Find the connection between the families of plants and flower diagrams, which are shown in the figure:

The codes of the answers:

- | | |
|---------------------------|-------------------------|
| 1. <i>Magnoliaceae</i> ; | 6. <i>Lamiaceae</i> ; |
| 2. <i>Ranunculaceae</i> ; | 7. <i>Solanaceae</i> ; |
| 3. <i>Rosaceae</i> ; | 8. <i>Liliaceae</i> ; |
| 4. <i>Fabaceae</i> ; | 9. <i>Orchidaceae</i> ; |
| 5. <i>Brassicaceae</i> ; | 10. <i>Poaceae</i> . |



Answers: a) b) c) d) e) f) g) h) i) j)

*****Following information deals with the questions № 170-172.
Solve following genetic problems.**

170. Black and white mice were hybridized. Hybrid individuals of the first generation were grey (agouti). After their hybridization segregation occurred in number of 9 grey, 3 black, 4 white. Find the genotypes of the parents and F_1 -cross and F_2 -cross. Write conventional letter signs into the shown below scheme.

p	black mouse x white mouse
F_1	grey mice
F_2	9 grey mice; 3 black mice; 4 white mice.

171. When brown-coloured spaniels of thoroughbred lines were crossed with white-coloured ones the resulting hybrid individuals occurred white-coloured. The hybrids of the second generation showed the segregation in proportion 12 white : 3 black : 1 brown. Find the genotypes of the parents, hybrids F_1 and F_2 . Write them into the shown below scheme.

p	brown spaniel x white spaniel
F_1	white spaniels
F_2	12 white spaniels; 3 black spaniels; 1 brown spaniel.

172. In cats the gene of black colour and the gene of red colour are connected with sex, located in the X-chromosome and produce incomplete dominance. While they are combined the turtle colour occurs. The turtle-coloured cat gave birth to 5 kittens in proportion 1 red : 2 turtle : 2 black. That red kitten was female. Find the genotype and the phenotype of the tom-cat sire, genotypes and sex of kittens. Write these data into the shown below scheme.

p	turtle-coloured cat x tom-cat
The phenotype:	
The genotype:	
F_1	1 red kitten; 2 turtle kittens; 2 black kittens.
Sex:	
The genotype:	

173. From the listed information collect the data about spider (*Arachnida*).

The codes of the answers:

1. Members:

- a) scorpion (*Scorpiones*); b) slater (*Oniscoidea*);
- c) crawfish (*Palinuridae*); d) cyclop (*Cyclopes*);
- e) mite (*Acarina*); f) solifuga (*Solifugae*).

2. Species number:

- a) nearly 1 million; b) nearly 35000; c) nearly 60000.

3. Their body is clearly subdivided into the head, thorax and abdomen.

4. The body is formed by cephalothorax - prosoma and abdomen - epistoma, can consist of different number of segments.

5. Segments are organized into tagmas - the head, the thorax, the abdomen.

In many species the basal of the head is covered by chitin fold - carapace.

6. Antennas are absent.

7. There is one pair of antennae.

8. There are two pairs of antennae.

9. There are mandibular and maxillae.

10. There are chelicerae and pedipalps.

11. There are 3 pairs of walking legs on the thorax.

12. There are 4 pairs of walking and feeding legs.

13. There are 5 pairs of walking legs.

14. The legs are uniramous.

15. The legs are biramous.

16. There are simple eyes (ocellus) in different number (from 1-2 pairs to 8 pairs; 4 pairs usually on the cephalothorax).

17. There are two types of eyes - compound (in the most species) or simple (ocellus).

18. The eyes are compound (in many species) or there is the unpaired naupliar eye.

19. The nervous system is formed alike the one in primitive Arthropods and even Annelids: this is the brain, consisting of oesophageal double ganglia and ventral nerve cord.

20. The nervous system is concentrated. The dorsal ganglia (brain) is fused with the subesophageal ganglia. In some species one or more nervous ganglia of ventral nerve cord are remained.

21. They are the unique invertebrate animals who are capable to distinguish sounds.

22. The excretory system is formed by Malpighian tubules.

23. Antennal glands or maxillary glands are excretory system.

24. Sexual glands and Malpighian tubules are excretory system.

25. There are two variants of respiratory system: tracheae and/or lungs.

Some small animals do not have neither lungs nor tracheae.

26. They breathe through the gills, but when they are absent - through the body surface.

27. Breathing occurs by the help of tracheae system or through the body surface.

28. There are respiratory pigments in the blood.

29. The process of development involves incomplete or complete metamorphosis.

30. In the most species the process of development involves metamorphosis. The certain larva - nauplius - hatches from the egg.

31. The process of development is a direct one or involves metamorphosis.

32. Moulting is common for larva as well as for adults.

33. Adults do not moult.